

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ME-404 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Thermal Engineering and Gas Dynamics
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. State the various types of cooling towers. Explain any one of them.

OR

Write short note on LMTD approach to design a heat exchanger.

2. Steam is supplied to a steam turbine at a pressure of 20 bar and 230°C. It is then expanded isentropically to a pressure of 1 bar. Determine the efficiency of plant and specific volume of steam at the end of expansion.
3. Compare various high pressure boiler like La mant and Benson Boilers and draw neat well nominclated diagram of a Loeffler Boiler.
4. What is a Heat Balance Sheet of a Boiler? Draw a Specimen Balance Sheet of a boiler with all loss and gain entries showing in it.
5. In a Rankine cycle, the steam enters the turbine is saturated at a pressure of 35 bar and the exhaust pressure is 0.2 bar. Determine:
- a) Pump work
 - b) Turbine work
 - c) Rankine cycle efficiency
 - d) Condenser heat flow and
 - e) Dryness at the end of expansion

Assume the flow rate of 9.5 kg/s

6. In a steam boiler 5400 kg of steam is produced per hour at a pressure of 7.5 bar with feed water at 41.5°C. The dryness fraction of steam at exit is 0.98. The amount of coal burnt per hour is 670 kg of calorific value 31000 kJ/kg. Determine:
- a) Boiler efficiency and
 - b) Equivalent Evaporation

OR

Deduce followings :

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2 + (v-1)M^2}{v+1} \right]^{\frac{v+1}{2(v-1)}} \text{ with usual notations.}$$

[2]

7. Steam at 50 bar, 400°C expands in a Rankine cycle to 0.34 bar. For a mass flow rate of 150kg/s of steam, determine :
- i) The power developed
 - ii) The thermal efficiency and
 - iii) Specific steam consumption

OR

Write short note on LMTD approach to design a heat exchanger.

8. Write short notes on any two :
- i) Super critical boiler
 - ii) Steam nozzle
 - iii) Classification of condensers
